



GIRRAWEE HIGH SCHOOL
YEAR 11 YEARLY EXAMINATION

2010

MATHEMATICS EXTENSION

*Time allowed - Two hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1, Question 2, etc. Write on only one side of the paper.
Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

Question1 (22 marks)

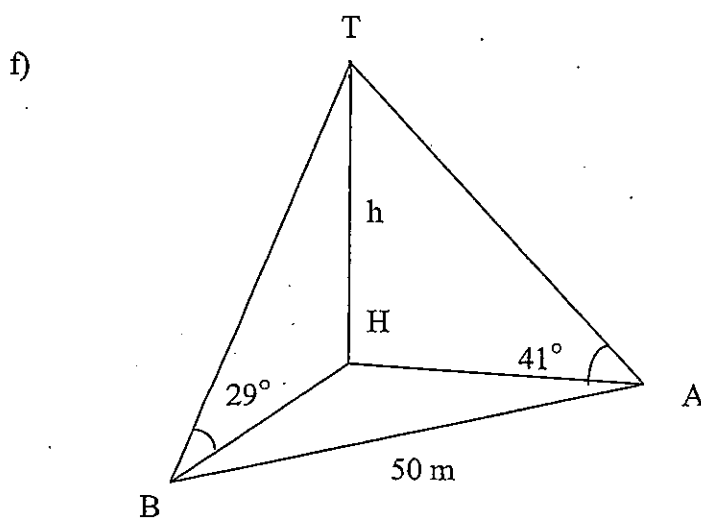
a) Solve: $\frac{2x+1}{x-2} \geq 1$ 3

b) Find the acute angle between the lines $x - 2y + 1 = 0$ and $y = 3x - 2$, correct to the nearest minute. 3

c) Find the coordinates of point P which divides the interval joining $A(-4, 2)$ and $B(2, -3)$ externally in the ratio 5:2. 3

d) Differentiate: $y = (2x - 1)^3 (x - 1)^2$ 3

e) Solve for x : $\frac{1}{\sqrt{3} + \sqrt{x}} + \frac{1}{\sqrt{3} - \sqrt{x}} = 2\sqrt{3}$ 3



The angle of elevation of a hill at a place A due east of it is 41° . At a place B due south of H the angle of elevation is 29° . If the distance from A to B is 50 metres and H is the foot of the hill,

(i) Show that $50^2 = h^2 (\tan^2 49^\circ + \tan^2 61^\circ)$ 4

(ii) Find the height of the hill (correct to 4 significant figures). 3

Question 2 (27 marks)

- a) Find the value of k if the remainder is 3 when $5x^2 - 10x + k$ is divided by $x - 1$. 2
- b) If $x = 1$ is a double root of $P(x) = 6x^4 - 7x^3 + cx^2 + 13x - 4$.
- (i) Show that $c = -8$ 1
- (ii) Hence, or otherwise, fully factorise 3
- $$P(x) = 6x^4 - 7x^3 - 8x^2 + 13x - 4$$
- (iii) Sketch the graph of $P(x) = 6x^4 - 7x^3 - 8x^2 + 13x - 4$ clearly showing the intercepts. 3
- (iv) Hence, or otherwise, solve $6x^4 - 7x^3 - 8x^2 + 13x - 4 \geq 0$ 2
- c) Find the value of k for which $kx^2 + 2kx + 3 = 0$ has roots that differ by 1. 3
- d) If α , β and γ are the roots of the equation $2x^3 - 5x^2 - 2x + 6 = 0$, find the values of:
- (i) $\alpha + \beta + \gamma$ 1
- (ii) $\alpha\beta + \beta\gamma + \alpha\gamma$ 1
- (iii) $\alpha\beta\gamma$ 1
- (iv) $\alpha^2 + \beta^2 + \gamma^2$ 3
- (v) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$ 3
- e) $P(x)$ is a polynomial of degree 3. When $P(x)$ is divided by $x - 2$, the remainder is 1. When $P(x)$ is divided by $x + 3$, the remainder is 2. When $P(x)$ is divided by $(x - 2)(x + 3)$, the remainder is $(ax + b)$. Find the values of a and b . 4

Question 3 (24 marks)

a) Given that $\cos x = \frac{8}{17}$ and $\tan y = \frac{5}{12}$ where x and y are acute, find the exact value of $\sin(x - y)$. 4

b) Simplify: $\frac{\sin 2A}{1 + \cos 2A}$ 3

c) Given that $t = \tan \frac{\theta}{2}$, express $\frac{1}{1 + \cos \theta - \sin \theta}$ in terms of t . 3

d) Solve for $0^\circ \leq \theta \leq 360^\circ$

(i) $\tan 2\theta = \frac{1}{\sqrt{3}}$ 3

(ii) $\tan^2 \theta + 5 \sec^2 \theta = 11$ 3

e) Express $\cos x + 7 \sin x$ in the form $A \cos(x + \alpha)$.
Hence, solve $\cos x + 7 \sin x = 5$, $0^\circ \leq x \leq 360^\circ$. 4

f) Solve $3 \sin \theta + \cos \theta = 2$ by using $t = \tan \frac{\theta}{2}$, $0^\circ \leq \theta \leq 360^\circ$. 4

Question 4 (23 marks)

- a) How many numbers greater than 500 can be formed from the digits 2, 3, 6 and 9 if each digit can only be used once? 3
- b) In how many ways can the letters of the word ISOSCELES be arranged using all the letters? 2
- c) In how many ways can the batting order of an 11 man cricket team be arranged if:
- (i) Monjee has to bat first? 2
 - (ii) Monjee has to bat first and Arnel has to bat last? 2
- d) A committee of 3 is to be formed from 4 boys and 5 girls. In how many ways can the committee be formed if:
- (i) the committee comprises all boys? 2
 - (ii) the committee contains 1 boy and 2 girls? 2
 - (iii) a particular girl will be on the committee? 2
- e) Seven people are to be seated at a round table.
- (i) How many seating arrangements are possible? 2
 - (ii) How many arrangements are possible if Clarissa and Rojda refuse to sit together? 3
- f) In how many ways can eight boys sit in an eight-oared boat (4 on each side) if 3 of the crew can only row on the stroke side and one particular boy must be on the bow side? 3

Question 5 (24 marks)

- a) Find the Cartesian equation of the parabola $x = 8t, y = 4t^2$. 3
- b) Find the parametric equations for the parabola $x^2 = 12y$ 3
- c) Consider the parabola $x^2 = 4ay$ where $a > 0$, and suppose the tangent at $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ intersect at the point T. Let $S(0, a)$ be the co-ordinates of the focus of the parabola.
- (i) Show that the equation of the chord PQ is 4
- $$y = \frac{1}{2}(p + q)x - apq$$
- (ii) Find the coordinates of M, the midpoint of PQ. 2
- (iii) If $\angle POQ = 90^\circ$, where O is the origin, prove that $pq = -4$. 3
- (iv) Show that the equation of the tangent at P is $y = px - ap^2$ 3
- (v) Show that the coordinates of T are $(a(p + q), apq)$ 3
- (vi) Show that $SP = a(p^2 + 1)$. 3

Question 1 (22 marks)

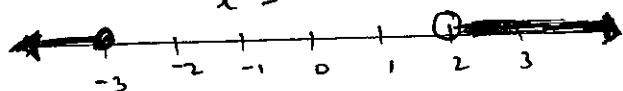
a) $\frac{2x+1}{x-2} \geq 1$

$x \neq 2$

Solve $\frac{2x+1}{x-2} = 1$

$2x+1 = x-2$

$x = -3$



Test: $x = -4$

$x = 0$

$x = 3$

$\frac{2(-4)+1}{-6} = \frac{-7}{-6} \geq 1$
✓

$\frac{1}{-2} \geq 1$
✗

$\frac{7}{1} \geq 1$
✓

∴ Solution: $x \leq -3, x > 2$

b) $x - 2y + 1 = 0$

$2y = x + 1$

$y = \frac{x}{2} + \frac{1}{2}$

$m_1 = \frac{1}{2}$

$y = 3x - 2$

$m_2 = 3$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{\frac{1}{2} - 3}{1 + \frac{1}{2} \times 3} \right|$

$= \left| \frac{-\frac{5}{2}}{\frac{5}{2}} \right| = 1$

$\tan \theta = 1$

$\theta = 45^\circ$

c) A (-4, 2) B (2, -3)

$x = \frac{2 \times -4 + -5 \times 2}{-3} = 6$

$y = \frac{2 \times 2 + -5 \times -3}{-3} = -6\frac{1}{3}$

P(6, -6 $\frac{1}{3}$)

d) $y = (2x-1)^3 (x-1)^2$

$\frac{dy}{dx} = v u' + u v'$

$= 6(2x-1)^2 (x-1)^2 +$
 $2(2x-1)^3 (x-1)$

$= 2(2x-1)^2 (x-1)(3x-3+2x-1)$

$= 2(2x-1)^2 (x-1)(5x-4)$

$u = (2x-1)$

$u' = 3(2x-1)$

$= 6(2x-1)$

$v = (x-1)^2$

$v' = 2(x-1)$

e) $\frac{1}{\sqrt{3} + \sqrt{5x}} + \frac{1}{\sqrt{3} - \sqrt{5x}} = 2\sqrt{3}$

$\frac{1}{\sqrt{3} + \sqrt{5x}} \times \frac{\sqrt{3} - \sqrt{5x}}{\sqrt{3} - \sqrt{5x}} + \frac{1}{\sqrt{3} - \sqrt{5x}} \times \frac{\sqrt{3} + \sqrt{5x}}{\sqrt{3} + \sqrt{5x}} = 2\sqrt{3}$

$\frac{\sqrt{3} - \sqrt{5x}}{3 - 5x} + \frac{\sqrt{3} + \sqrt{5x}}{3 - 5x} = 2\sqrt{3}$

$\frac{2\sqrt{3}}{3 - 5x} = 2\sqrt{3}$

$3 - 5x = 1$

$5x = 2$

f) i) $\angle ATH = 49^\circ, \angle BTH = 61^\circ$

In $\triangle ATH$,

$\tan 49^\circ = \frac{AH}{h}$

$AH = h \tan 49^\circ$

In $\triangle BTH$

$\tan 61^\circ = \frac{BH}{h}$

$BH = h \tan 61^\circ$

In $\triangle BHA$, using Pythagoras' theorem

$BA^2 = AH^2 + BH^2$

$50^2 = h^2 \tan^2 49^\circ + h^2 \tan^2 61^\circ$

$50^2 = h^2 (\tan^2 49^\circ + \tan^2 61^\circ)$

ii) $h = \frac{50}{\sqrt{\tan^2 49^\circ + \tan^2 61^\circ}}$

$= 23.37 \text{ m}$

1 mark for
4 sig figs

Question 2 (27 marks)

a) $P(1) = 5(1)^2 - 10(1) + k = 3$

$$5 - 10 + k = 3$$

$$k = 8 \quad (2)$$

b) $P(x) = 6x^4 - 7x^3 + cx^2 + 13x - 4$

i) $P(1) = 6 - 7 + c + 13 - 4 = 0$

$$c = -8 \quad (1)$$

ii)

$$\begin{array}{r} 6x^2 + 5x - 4 \\ x^2 - 2x + 1 \overline{) 6x^4 - 7x^3 - 8x^2 + 13x - 4} \\ \underline{-6x^4 + 12x^3 - 6x^2} \\ 5x^3 - 14x^2 + 13x \\ \underline{-5x^3 + 10x^2 - 5x} \\ -4x^2 + 8x - 4 \\ \underline{-4x^2 + 8x - 4} \\ 0 \end{array}$$

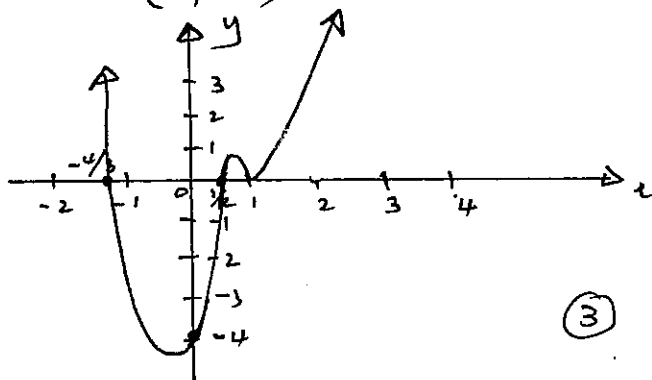
$$\therefore P(x) = (x-1)^2 (6x^2 + 5x - 4)$$

$$= (x-1)^2 (3x+4)(2x-1) \quad (3)$$

iii) $x \text{ int} \Rightarrow y = 0$

$(1, 0)$, $(-\frac{4}{3}, 0)$, $(\frac{1}{2}, 0)$
double

$y \text{ int} \Rightarrow x = 0$
 $(0, -4)$



iv) $P(x) \geq 0$ when

$$x \leq -\frac{4}{3}, x \geq \frac{1}{2} \quad (2)$$

c) $kx^3 + 2kx + 3 = 0$

roots : $\alpha, \alpha+1$

Sum : $2\alpha + 1 = -2$

$$2\alpha = -3$$

$$\alpha = -\frac{3}{2}$$

Prod : $\alpha^2 + \alpha = \frac{3}{k}$

$$\left(-\frac{3}{2}\right)^2 + \left(-\frac{3}{2}\right) = \frac{3}{k}$$

$$\frac{9}{4} - \frac{3}{2} = \frac{3}{k}$$

$$\frac{3}{4} = \frac{3}{k}$$

$$k = 4 \quad (3)$$

d) $2x^3 - 5x^2 - 2x + 6 = 0$

i) $\alpha + \beta + \gamma = \frac{5}{2} \quad (1)$

$$(1)$$

ii) $\alpha\beta + \beta\gamma + \alpha\gamma = -1 \quad (1)$

$$(1)$$

iii) $\alpha\beta\gamma = -3$

iv) $\alpha^2 + \beta^2 + \gamma^2$

$$= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$$

$$= \left(\frac{5}{2}\right)^2 - 2(-1)$$

$$= \frac{25}{4} + 2$$

$$= 8\frac{1}{4} \quad (3)$$

v) $\frac{1}{\alpha^2\beta^2} + \frac{1}{\beta^2\gamma^2} + \frac{1}{\alpha^2\gamma^2}$

$$= \frac{\gamma^2 + \alpha^2 + \beta^2}{(\alpha\beta\gamma)^2} = \frac{\frac{33}{4}}{(-3)^2}$$

$$= \frac{11}{12} \quad (3)$$

e) $P(x) = (x-2)(x+3)Q(x) + ax + b$

$$P(2) = 2a + b = 1 \quad (1)$$

$$P(-3) = -3a + b = 2 \quad (2)$$

$$(1) - (2)$$

$$5a = -1$$

$$a = -\frac{1}{5}$$

$$2\left(-\frac{1}{5}\right) + b = 1$$

$$b = \frac{7}{5} \quad (4)$$

$$a = -\frac{1}{5}, b = \frac{7}{5}$$

Question 3 (24 marks)

a) $\cos x = \frac{8}{17}$; $\tan y = \frac{5}{12}$

$$\begin{aligned}\sin(x-y) &= \sin x \cos y - \cos x \sin y \\ &= \frac{15}{17} \cdot \frac{12}{13} - \frac{8}{17} \cdot \frac{5}{13} \\ &= \frac{140}{221}\end{aligned}$$

(4)

b) $\frac{\sin 2A}{1 + \cos 2A} = \frac{2 \sin A \cos A}{1 + (1 - 2 \sin^2 A)}$

$$\begin{aligned}&= \frac{2 \sin A \cos A}{2 - 2 \sin^2 A} = \frac{2 \sin A \cos A}{2(1 - \sin^2 A)} \\ &= \frac{\sin A \cos A}{\cos^2 A} = \frac{\sin A}{\cos A} \\ &= \tan A\end{aligned}$$

(3)

c) $\frac{1}{1 + \cos \theta - \sin \theta}$

$$\begin{aligned}&= \frac{1}{1 + \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2}} \\ &= \frac{1}{\frac{1+t^2+1-t^2-2t}{1+t^2}} \\ &= \frac{1}{\frac{2-2t}{1+t^2}} = 1 \times \frac{1+t^2}{2-2t} \\ &= \frac{1+t^2}{2(1-t)}\end{aligned}$$

(3)

d) (i) $\tan 2\theta = \frac{1}{\sqrt{3}}$ $\phi 1, 3$

$$\begin{aligned}0^\circ \leq 2\theta \leq 720^\circ \\ 2\theta &= 30^\circ, 210^\circ, 390^\circ, 570^\circ \\ \theta &= 15^\circ, 105^\circ, 195^\circ, 285^\circ\end{aligned}$$

(3)

d) $\tan^2 \theta + 5 \sec^2 \theta = 11$

$$\tan^2 \theta + 5(1 + \tan^2 \theta) = 11$$

$$\tan^2 \theta + 5 + 5 \tan^2 \theta = 11$$

$$6 \tan^2 \theta = 6$$

$$\tan^2 \theta = 1$$

$$\tan \theta = \pm 1$$

$$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$

(3)

e) $\cos x + 7 \sin x = A \cos(x-\alpha)$

$$A = \sqrt{50}$$

$$\frac{1}{\sqrt{50}} \cos x + \frac{7}{\sqrt{50}} \sin x = \cos(x-\alpha)$$

$$= \cos x \cos \alpha + \sin x \sin \alpha$$

$$\cos \alpha = \frac{1}{\sqrt{50}} ; \sin \alpha = \frac{7}{\sqrt{50}}$$

$$\alpha < 90^\circ$$

$$\tan \alpha = 7$$

$$\alpha = 81^\circ 52'$$

$$\therefore \cos x + 7 \sin x = \sqrt{50} \cos(x + 81^\circ 52')$$

$$\sqrt{50} \cos(x + 81^\circ 52') = 5$$

$$\cos(x + 81^\circ 52') = \frac{1}{\sqrt{2}}$$

$$x + 81^\circ 52' = 45^\circ, 315^\circ, 405^\circ$$

$$x = 233^\circ 08', 323^\circ 08'$$

(4)

f) $3 \sin \theta + \cos \theta = 2$

$$3\left(\frac{2t}{1+t^2}\right) + \frac{1-t^2}{1+t^2} = 2$$

$$6t + 1 - t^2 = 2 + 2t^2$$

$$3t^2 - 6t + 1 = 0$$

$$t = \frac{6 \pm \sqrt{24}}{6}$$

$$\tan \frac{\theta}{2} = \frac{3 \pm \sqrt{6}}{3} \quad 0^\circ \leq \frac{\theta}{2} \leq 15^\circ$$

$$\frac{\theta}{2} = 10^\circ 24', 61^\circ 10'$$

$$\theta = 20^\circ 48', 122^\circ 20'$$

Test $\theta = 180^\circ$

$$3 \sin 180^\circ + \cos 180^\circ = -1 \neq 2$$

$$\therefore \text{not a solution}$$

(4)

Solution: $\theta = 20^\circ 48', 122^\circ 20'$

Question 4 (23 marks)

a) 3 digit nos $\Rightarrow 2 \times 3 \times 2 = 12$
4 digit nos $\Rightarrow 4 \times 3 \times 2 = 24$
 $\therefore \text{Total} = 36$ (3)

b) ISOSCELES $\begin{matrix} 9 \\ 3S \\ 2E \end{matrix}$

$\frac{9!}{3!2!} = 30240$ (2)

c) i) $10! = 3628800$ (2)

ii) $9! = 362880$ (2)

d) 4B 5G

i) $3B : {}^4C_3 = 4$ (2)

ii) $1B2G : {}^4C_1 \times {}^5C_2 = 40$ (2)

iii) ${}^8C_2 = 28$ (2)

e) i) $6! = 720$ (2)

ii) not together
 $= 6! - \text{together}$
 $= 6! - 2 \times 5!$
 $= 720 - 240$
 $= 480$ (3)

f) Stroke side : $4 \times 3!$ $\swarrow 3B$
Bow side : $4 \leftarrow 1B$
Remainder = $4!$

$\therefore 4 \times 3! \times 4 \times 4!$
 $= 2304$ (3)

Question 5 (24 marks)

a) $x = 8t$; $y = 4t^2$

$$t = \frac{x}{8}$$

$$y = 4 \left(\frac{x}{8} \right)^2$$

$$= \frac{4x^2}{64} = \frac{x^2}{16}$$

$$x^2 = 16y$$

(3)

b) $x^2 = 12y$

$$4a = 12$$

$$a = 3$$

$$\therefore x = 2at \text{ ; } y = at^2$$

$$x = 6t \text{ ; } y = 3t^2$$

(3)

c) $x^2 = 4ay$

i) $P(2ap, ap^2)$ $Q(2aq, aq^2)$

$$m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p+q)(p-q)}{2a(p-q)}$$

$$m = \frac{p+q}{2}$$

$E_{PQ} : y - ap^2 = \frac{p+q}{2}(x - 2ap)$

$$2y - 2ap^2 = px + qx - 2ap^2 - 2apq$$

$$2y = (p+q)x - 2apq$$

$$y = \frac{1}{2}(p+q)x - apq$$

(4)

ii) $M = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$

$$= \left(\frac{2a(p+q)}{2}, \frac{a(p^2+q^2)}{2} \right)$$

$$= \left(a(p+q), \frac{a}{2}(p^2+q^2) \right)$$

(2)

iii) $\angle POQ = 90^\circ$

$$\therefore m_{OP} \cdot m_{OQ} = -1$$

$$m_{OP} = \frac{ap^2}{2ap} = \frac{p}{2}$$

$$m_{OQ} = \frac{aq^2}{2aq} = \frac{q}{2}$$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4$$

(3)

iv) $m_{\text{tangent at } P} = p$ $P(2ap, ap^2)$

$$E : y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

(3)

v) $E_{\text{tangent at } P} : y = px - ap^2$ — (1)

$$E_{\text{tangent at } Q} : y = qx - aq^2$$
 — (2)

(1) - (2) $0 = (p-q)x - (p^2 - q^2)a$

$$(p-q)x = (p+q)(p-q)a$$

$$x = a(p+q)$$

Substitute $x = a(p+q)$ into (1)

$$y = p \cdot a(p+q) - ap^2$$

$$y = ap^2 + apq - ap^2$$

$$y = apq$$

(3)

$$\therefore T = (a(p+q), apq)$$

vi) $S(0, a)$ $P(2ap, ap^2)$

$$SP = \sqrt{(2ap)^2 + (ap^2 - a)^2}$$

$$= \sqrt{4a^2p^2 + a(p^2 - 1)^2}$$

$$= \sqrt{4a^2p^2 + a^2(p^4 - 2p^2 + 1)}$$

$$= \sqrt{4a^2p^2 + a^2p^4 - 2a^2p^2 + a^2}$$

$$= \sqrt{a^2p^4 + 2a^2p^2 + a^2}$$

$$= \sqrt{a^2(p^2 + 1)^2} = a(p^2 + 1)$$

(3)

